

# TOAST Stroke Classification

Trial of Org 10172 in Acute Stroke Treatment (TOAST) diagnostic criteria for ischemic stroke etiology.

## ACUTE ISCHEMIC STROKE

Large Artery (LAA)

Small Vessel (SVO)

Cardioembolic (CE)

Other Det. (ODE)

Undetermined Etiology

### 1. Large-Artery Atherosclerosis (LAA)

- **Clinical:** Cortical signs (aphasia, neglect, gaze deviation) or brainstem/cerebellar syndrome.
- **Imaging:** Cortical/subcortical infarct  $> 1.5$  cm on MRI or CT.
- **Vascular:**  $> 50\%$  stenosis or occlusion of the relevant major extracranial (carotid, vertebral) or intracranial artery.
- **Exclusion:** Must exclude a high-risk cardioembolic source.

### 2. Small-Vessel Occlusion (SVO / Lacune)

- **Clinical:** Classic lacunar syndrome (pure motor, pure sensory, sensorimotor, ataxic hemiparesis, clumsy hand) **WITHOUT** cortical signs.
- **Imaging:** Normal scan or deep subcortical lesion  $\leq 1.5$  cm.
- **Vascular/Cardiac:** Relevant artery must lack  $> 50\%$  stenosis, and patient must lack high-risk cardioembolic sources.

### 4. Other Determined Etiology (ODE)

- **Clinical/Imaging:** Infarction of any size with diagnostic proof of a rare/specific underlying mechanism:
- Arterial dissection (e.g. carotid or vertebral dissection)
- CNS vasculitis or systemic vasculopathy
- RCVS (Reversible Cerebral Vasoconstriction Syndrome)
- Moya-Moya disease, CADASIL, or Fibromuscular Displasia
- Prothrombotic/hypercoagulable state (APLS, cancer, DIC)

### 3. Cardioembolism (CE)

#### HIGH-RISK SOURCES

- Atrial Fibrillation / Flutter
- Mechanical prosthetic valve
- Left atrial / LAA thrombus
- Recent anterior MI ( $< 3$  mo)
- Dilated cardiomyopathy (EF  $< 30\%$ )
- Infective endocarditis
- Sick sinus syndrome / LA myxoma

#### MEDIUM-RISK SOURCES

- PFO + Atrial Septal Aneurysm
- Mitral valve prolapse
- Mitral annulus calcification
- Bioprosthetic heart valve
- Calcific aortic stenosis
- LV dysfunction (EF 30–40%)
- LA spontaneous echo contrast

### 5. Undetermined Etiology

- **Due to competing risks:**  $\geq 2$  potential etiologies found (e.g., active AFib AND  $\geq 50\%$  ipsilateral carotid stenosis).
- **Negative evaluation:** Complete diagnostic workup identifies no clear source (Cryptogenic stroke).
- **Incomplete evaluation:** Workup is unfinished (e.g., patient discharged/AMA before Echo or vascular imaging).

**ESUS Criteria:** non-lacunar stroke, no relevant  $> 50\%$  stenosis, no high-risk cardioembolic source, negative ECG/telemetry  $\geq 24$  hours.

## REQUIRED DIAGNOSTIC WORKUP TO COMPLETE TOAST CLASSIFICATION

- |                                                                                                               |                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> <b>Parenchymal:</b> MRI Brain (DWI/ADC) preferred, or CT Head.            | <input checked="" type="checkbox"/> <b>Vascular:</b> CTA or MRA Head & Neck (or Carotid Duplex + TCD).          |
| <input checked="" type="checkbox"/> <b>Rhythm:</b> EKG + Continuous Telemetry $\geq 24$ h (or loop recorder). | <input checked="" type="checkbox"/> <b>Cardiac:</b> TTE required; consider TEE if cryptogenic / ESUS suspected. |

**Original Study:** Adams HP Jr, et al. TOAST. *Stroke*. 1993;24:35-41. | **AHA/ASA Guideline:** Kleindorfer DO, et al. 2021 Stroke Prevention. *Stroke*. 2021;52:e364-e467.